Algebra Review: Unit 8 — Quadratics

Answer Key

Station 1: Solving by Factoring and Square Roots

1. $x = -3, x = -2$
2. $x = 3, x = 4$
3. $x = -4, x = 5$
4. $x = 0, x = 4$
5. $x = -4, x = 4$
6. $x = \pm 2\sqrt{5}$
7. $x = -2, x = 2$
8. $x = -2, x = 8$
9. $x = -2, x = \dfrac{3}{2}$
10. $x = -1 \pm 3\sqrt{2}$

Station 2: Completing the Square

1. 16
2. 25
3. $\dfrac{9}{4}$
4. $x = -7,\ x = 1 \quad (x+3)^2 = 16$
5. $x = 1,\ x = 7 \quad (x-4)^2 = 9$
6. $x = -2 \pm \sqrt{5} \quad (x+2)^2 = 5$
7. $x = 1 \pm \sqrt{6} \quad (x-1)^2 = 6$
8. $x = -3,\ x = 2 \quad \left(x + \frac{1}{2}\right)^2 = \dfrac{25}{4}$
9. $x = -7,\ x = 1 \quad$ *divide by 2 first*
10. $y = (x-3)^2 - 4$

Station 3: The Quadratic Formula and the Discriminant

1. $b^2 - 4ac = 49$; two real solutions
2. $b^2 - 4ac = 0$; one real solution
3. $b^2 - 4ac = -16$; no real solutions
4. $x = -2, x = 5$
5. $x = -3, x = \dfrac{1}{2}$
6. $x = -3 \pm \sqrt{5}$
7. $x = 2 \pm \sqrt{3}$
8. $x = \dfrac{1 \pm \sqrt{7}}{3}$
9. $x = \dfrac{1}{2}$
10. $x = \dfrac{3 \pm \sqrt{3}}{2}$

Station 4: Graphs of Quadratic Functions and Applications

1. $(2, -1)$; $x = 2$; opens up; y-int 3
2. $(3, 4)$; $x = 3$; opens down; y-int -5
3. $(-2, -3)$; $x = -2$; opens up; y-int 5
4. $(1, -4)$; $x = -1,\ x = 3 \quad (x-1)^2 = 4$
5. $(-3, 8)$; $x = -5,\ x = -1$; opens down $\quad (x+3)^2 = 4$
6. (a) $(-1, 4)$ (b) $x = -1$ (c) $x = -3,\ x = 1$ (d) $(0, 3)$
7. $(2, 5)$
8. Maximum height 64 ft at $t = 1$ s
9. $t = 3$ seconds $\quad -16(t-3)(t+1) = 0$
10. Price \$10; revenue \$500